

Capsule Endoscopy Image Classification using ResNeXt & VCS-Optimized KNN with XAI

Abstract—Gastrointestinal conditions such as arteriovenous malformations (AVM) and ulcers require timely and accurate detection, yet manual capsule endoscopy review is time-consuming and prone to variability. This study presents a hybrid framework combining ResNeXt-50 for deep feature extraction with a K-Nearest Neighbors classifier optimized by the Virus Colony Search (VCS) metaheuristic. The approach enhances accuracy and robustness through adaptive decision boundaries and efficient optimization. Experiments on a balanced dataset of 9,000 images (AVM, Ulcer, Normal) achieved 96.15% validation accuracy, Cohen’s Kappa of 0.939, and a Macro F1-score of 0.95, outperforming CNN, Transformer, and traditional baselines. Grad-CAM, Grad-CAM++, and LIME visualizations confirmed the model’s focus on clinically relevant regions, supporting transparency and reliability for AI-assisted gastrointestinal diagnosis.

Index Terms—Capsule Endoscopy, Gastrointestinal Disease, Ulcer Detection, Virus Colony Search (VCS), ResNeXt-50, KNN, Explainable AI (XAI).

I. INTRODUCTION

Wireless capsule endoscopy (WCE) has revolutionized gastrointestinal (GI) diagnostics by enabling non-invasive, full-length visualization of the small intestine, a region traditionally difficult to access with conventional endoscopy [1]. As a self-contained capsule capturing thousands of frames during transit, WCE provides broad diagnostic coverage for conditions such as arteriovenous malformations (AVMs), ulcers, and neoplasms. However, the manual review of over 50,000 frames per patient is time-consuming, labor-intensive, and prone to errors from fatigue and subtle lesion appearance.

With the growing clinical use of WCE, the large volume of visual data has placed significant pressure on diagnostic workflows. A single study requires 45–60 minutes of review, often exceeding an hour in complex cases, and error rates can reach 10–20% for subtle lesions such as small erosions or vascular malformations. Artificial intelligence has emerged as an effective aid, with deep learning models surpassing 90% accuracy for bleeding and polyp detection [2]. For example, convolutional neural networks (CNNs) trained on dedicated datasets have achieved 92% sensitivity for polyps and 94% for bleeding. Despite these advances, CNNs remain limited by class imbalance, as rare but critical findings like tumors often represent less than 5% of images, leading to misclassification in practice.

Convolutional neural networks also face architectural constraints [3]. Their reliance on local receptive fields limits their ability to capture global spatial dependencies and contextual semantics which are often vital in differentiating between visually similar conditions such as ulcers and superficial

erosions. Moreover while CNNs excel in high-throughput image processing they are often criticized for their limited interpretability. For clinicians to trust automated systems the reasoning behind predictions must be transparent and aligned with clinical decision-making. In response to these issues recent studies have incorporated attention mechanisms or class-weighting strategies to improve learning from imbalanced datasets [4]. Although these techniques have contributed to incremental progress, they have not fully addressed the pressing need for models that simultaneously provide precision robustness and interpretability in diverse patient populations.

Artificial intelligence offers a promising solution to the challenges of wireless capsule endoscopy by improving accuracy, efficiency, and reliability in image interpretation. Deep learning models can process thousands of frames quickly, detect subtle lesions, and distinguish between visually similar conditions that are often overlooked by human observers. Class balancing strategies and transfer learning further enhance detection of rare but critical findings such as small tumors, while explainable AI tools provide transparency and build clinician trust [5]. Automated triage systems also help reduce review time from hours to minutes, lowering fatigue related errors. By combining precision with interpretability, artificial intelligence has the potential to make capsule endoscopy more effective and scalable, ultimately improving gastrointestinal disease diagnosis and patient outcomes.

- Developed a hybrid framework using ResNeXt-50 feature extraction with a K-Nearest Neighbors classifier, optimized by the Virus Colony Search (VCS) metaheuristic.
- Employed the Capsule Endoscopy (KAUHC) dataset of 3,301 images, augmented to 9,000 balanced samples across three classes, with a 70/15/15 split.
- Achieved 96.15% accuracy with strong reliability metrics (Kappa = 0.939, MCC = 0.940, Macro F1 = 0.95), surpassing CNN, Transformer, and traditional baselines.
- Enhanced interpretability via Grad-CAM, Grad-CAM++, and LIME, consistently highlighting clinically relevant lesion regions.

The paper is organized as follows: Section I introduces the topic; Section II reviews related work; Section III describes the proposed method; Section IV presents the results and analysis; and Section V concludes with key findings.

II. LITERATURE REVIEW

Recent research on wireless capsule endoscopy (WCE) highlights advances in automated lesion detection, mucosal

visualization, and diagnostic accuracy, as summarized in Table VII. Sahafi et al. [6] employed YOLO-V8 variants for polypoid lesion segmentation on the KID dataset, achieving a Dice score of 93.99% but constrained by limited sample diversity. Wang et al. [7] introduced the Spectrum Aided Vision Enhancer (SAVE) using hyperspectral imaging on 526 MACE images, improving mucosal visualization with an SSIM of 91%, though validation was limited to a single center with mostly mild GERD cases. Bai et al. [8] proposed a transformer with spatial pooling for disease classification, reporting 79.15% accuracy on the multi-class Kvasir-Capsule dataset, underscoring challenges of small, imbalanced datasets despite strong binary performance on the Red Lesion Endoscopy dataset. Varam et al. [9] tested deep learning classifiers, with Vision Transformer achieving up to 97% F1-score on a balanced subset of Kvasir-Capsule, but emphasized that explainable AI integration remains at an experimental stage without clinical deployment. Finally, Kim and Chun [10] evaluated magnetically controlled capsule endoscopy in 350 patients, reporting 93.4% diagnostic accuracy for gastric lesions, though requiring specialized hardware and raising concerns about generalizability. Collectively, these studies illustrate the promise of AI-enhanced WCE across segmentation, classification, and visualization, while highlighting limitations in dataset scale, external validation, and clinical translation.

TABLE I: Concise Comparative Analysis of Recent Studies

Author	Method	Dataset	Results	Limitation
Sahafi et al. (2024) [6]	YOLO-V8	KID WCE	93.9%	Limited diversity/validation
Wang et al. (2024) [7]	Spectrum Aided Vision Enhancer	MACE	91% (SSIM)	Single-center, mild cases
Bai et al. (2022) [8]	Transformer (spatial pooling)	Kvasir-Capsule	79.1%	Small, imbalanced datasets
Varam et al. (2023) [9]	ViT	Kvasir-Capsule (subset)	97%	Subset; XAI experimental
Kim & Chun (2021) [10]	MCE	350 patients	93.4%	Requires MCE hardware

III. METHODOLOGY

The research began with an extensive review of the relevant literature to establish precise objectives. Based on these findings, the dataset and models were developed, assessed, and benchmarked against existing approaches to determine the most effective solution.

A. Dataset Description

Accurate detection of abnormalities in capsule endoscopy images is essential for reliable gastrointestinal disease diagnosis. This study employs the Capsule Endoscopy Dataset (KAUHC) [11], sourced from Kaggle. The dataset contains 3,301 endoscopic images organized into three classes: AVM (673), Ulcer (472), and Normal (2,156). Although the dataset

is moderately imbalanced, the inclusion of both pathological and normal categories offers a practical foundation for developing and evaluating deep learning models in gastrointestinal image classification research.

B. Data preprocessing

All images were resized to $224 \times 224 \times 3$ for model compatibility. Preprocessing included CLAHE for contrast enhancement, high-boost filtering for sharpening, and texture-edge filters such as Gabor, Sobel, and LoG, which were stacked to enrich feature representation [12]. To mitigate class imbalance, data augmentation (flips, rotations, scaling, translations, Gaussian noise, and brightness-contrast shifts) expanded the dataset to 9,000 balanced samples (3,000 per class). Sample outputs are shown in Fig. 1.

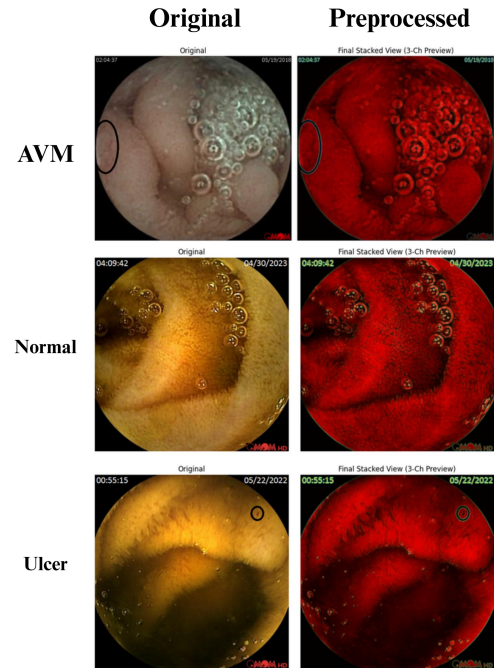


Fig. 1: Samples of preprocessed images

C. Dataset Split

A split of 70% training, 15% validation, and 15% testing was applied to the dataset, ensuring balanced learning and evaluation. Table II presents the distribution across classes.

TABLE II: Dataset Distribution After Augmentation

Class	Training	Validation	Testing	Total
AVM	2100	450	450	3000
Ulcer	2100	450	450	3000
Normal	2100	450	450	3000
Total	6300	1350	1350	9000

D. Proposed model architecture

The proposed model employs a ResNeXt-50 backbone [13] as the primary feature extractor, leveraging its aggregated

residual transformations to capture both local patterns and high-level semantic information from endoscopic images. The original classification head of ResNeXt is replaced with a lightweight normalization and dropout block (rate = 0.5) applied after feature extraction to reduce overfitting and produce stable embeddings. This stage outputs compact, L2-normalized, and discriminative feature vectors suitable for downstream classification.

Rather than relying solely on deep learning’s fully connected layer for classification, the normalized embeddings are further processed using a K-Nearest Neighbors (KNN) classifier [14]. To ensure optimal decision boundaries, the KNN hyperparameters, including the number of neighbors and the Minkowski distance metric, are automatically tuned through the Virus Colony Search (VCS) optimization algorithm [15]. This hybrid design combines the representational strength of ResNeXt features with the flexibility of an optimized distance-based classifier.

By integrating deep feature extraction via metaheuristic optimization with KNN classification, the model achieves a balance between robust representation and adaptive decision-making. This pipeline provides a scalable and interpretable framework for capsule endoscopy image classification. The architecture of the proposed model is illustrated in Fig. 2.

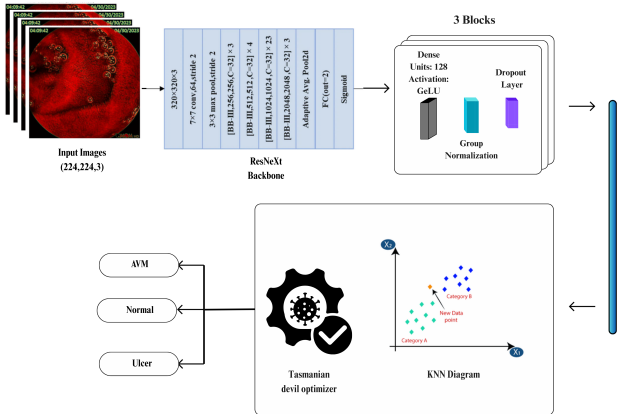


Fig. 2: Proposed Model Architecture

E. Hyperparameters

Cross-Entropy loss and the AdamW optimizer were used for training, with a ReduceLROnPlateau scheduler and dropout (0.5) to enhance generalization. Early stopping (patience = 3) typically led to convergence within 25–30 epochs. Mixed-precision training with gradient accumulation provided an effective batch size of 64.

Training was performed on a Google Colab GPU. After training the ResNeXt-50 feature extractor, L2-normalized embeddings were classified using KNN with Minkowski distance (Eq. 1). The number of neighbors and distance order $p \in \{1, 2\}$ were optimized via Virus Colony Search (VCS), which uses bounded mutation (Eq. 2) and guided recombination (Eq. 3).

VCS ran with a population of 20 for 10 iterations. The final hyperparameters are summarized in Table III.

$$d_p(\mathbf{u}, \mathbf{v}) = \left(\sum_{j=1}^d |u_j - v_j|^p \right)^{1/p} \quad (1)$$

$$\mathbf{x}^{\text{mut}} = \text{clip!}(\mathbf{x} + \epsilon, \ell, \mathbf{u}), \quad \epsilon \sim \mathcal{U}(-s, s)^D \quad (2)$$

$$\mathbf{x}^{\text{rec}} = \text{clip!}(\mathbf{x}^{\text{mut}} + \alpha \odot (\mathbf{x}^* - \mathbf{x}^{\text{mut}}), \ell, \mathbf{u}), \quad \alpha \sim \mathcal{U}(-r, r)^D \quad (3)$$

TABLE III: Hyperparameters for ResNeXt–KNN Hybrid Model

Training Hyperparameters	
Parameter	Value
Batch Size	32 (Effective 64 with accumulation)
Number of Epochs	100
Learning Rate	1×10^{-5}
Optimizer	AdamW
Loss Function	CrossEntropyLoss
Early Stopping Patience	3
Model Architecture Parameters	
Parameter	Value
Image Size	224×224
Backbone Model	ResNeXt-50 ($32 \times 4d$)
Feature Dimension	2048
Dropout Rate	0.5
Classifier	K-Nearest Neighbors (KNN)
Distance Metric	Minkowski, $p \in \{1, 2\}$
Optimization Parameters (VCS)	
Parameter	Value
Population Size	20
Iterations	10
Optimized Params	K (neighbors), p (distance order)

F. Model Evaluation Matrices

Model performance was evaluated using accuracy, F1-score, Cohen’s Kappa, MCC, specificity, FMI, and CSI. Training–validation curves and the confusion matrix assessed learning behavior and class-wise predictions. ROC, Precision–Recall, cumulative gain, lift, calibration, and entropy plots further verified prediction reliability under class imbalance [16].

G. Explainable AI Methods

LIME (Eqn. 4) improves model transparency by generating local surrogate models that approximate the predictions of complex classifiers [17]. Grad-CAM (Eqn. 5) and its extension Grad-CAM++ (Eqn. 6) enhance interpretability by visualizing class-discriminative regions in input images through gradient-based feature importance [18], [19].

$$\xi(x) = \arg \min_{g \in G} \sum_i \pi_x(z_i) (f(z_i) - g(z_i))^2 + \Omega(g) \quad (4)$$

$$L_{\text{Grad-CAM}}^c = \text{GeLU} \left(\sum_k \alpha_k^c A^k \right) \quad (5)$$

$$L_{\text{Grad-CAM}^{++}}^c = \text{GeLU} \left(\sum_k \alpha_k^{c++} A^k \right) \quad (6)$$

IV. RESULTS AND ANALYSIS

The results section is structured into three main parts: model selection, assessment of performance metrics, and visualization of outcomes using charts and plots.

A. Proposed Model Matrix Analysis

The proposed model achieves strong performance with 96.15% accuracy and a Cohen's Kappa of 0.939 (Table IV). Macro-averaged precision, recall, and F1-score are 93%, 95%, and 95%, reflecting balanced classification across all categories. The AVM class attained 96% precision and 92% recall, while Normal achieved 92% precision and 98% recall. Similarly, the Ulcer class reached 96% precision and 94% recall. With an overall specificity of 97.5%, the model demonstrates robust performance and reliable diagnostic capability.

TABLE IV: Per-Class Performance Metrics

Class	Acc.	Prec.	Rec.	F1	Spec.
AVM	0.957	0.96	0.92	0.94	0.969
Normal	0.979	0.92	0.98	0.95	0.986
Ulcer	0.962	0.96	0.94	0.95	0.969
Overall	0.9615	0.93	0.95	0.95	0.975

Class	MCC	Kappa	FMI	CSI
AVM	0.932	0.931	0.950	0.918
Normal	0.949	0.948	0.955	0.940
Ulcer	0.938	0.937	0.952	0.928
Overall	0.940	0.939	0.952	0.929

B. Ablation Study

An ablation study (Table V) evaluates the contribution of each component in the ResNeXt–KNN model. The complete model achieved the highest accuracy of 96.15%, demonstrating the advantage of combining ResNeXt feature extraction with KNN classification optimized through VCS. Removing ResNeXt dropped accuracy to 78.42%, while excluding KNN or VCS reduced it to 94.10% and 95.02%, respectively. The classifier alone performed the worst at 53.87%, highlighting the importance of deep features and optimization.

TABLE V: Ablation Study of the ResNeXt–KNN Model

Variant	ResNeXt	KNN	VCS	Acc.	Params (M)
Full Model	✓	✓	✓	96.15%	25.0
Without ResNeXt	–	✓	✓	78.42%	0.1
Without KNN	✓	–	–	94.10%	25.0
Without VCS	✓	✓	–	95.02%	25.0
Only Classifier	–	✓	–	53.87%	0.1

C. Comparison of Models, Classifiers, and Optimizers

Table VI shows that the proposed ResNeXt–KNN with VCS achieved the best performance with 96.15% accuracy, surpassing the baseline KNN classifier (94.50%) and the ResNeXt50 feature extractor (94.10%). These results demonstrate the effectiveness of combining deep ResNeXt features with KNN classification optimized through VCS.

TABLE VI: Performance Comparison of the Proposed Model

Model	Acc.	Kappa	MCC	FMI	CSI
Classifier Models					
KNN (Baseline)	94.50%	0.940	0.938	0.945	0.932
DenseNet121	93.10%	0.927	0.923	0.931	0.920
Xception	92.85%	0.924	0.920	0.928	0.915
DT (Handcrafted)	89.72%	0.892	0.890	0.901	0.885
Feature Extractor Models					
ResNeXt50	94.10%	0.936	0.934	0.941	0.928
CaiT	94.22%	0.942	0.939	0.947	0.933
PVT	93.67%	0.938	0.934	0.941	0.926
Proposed Model					
ResNeXt–KNN (VCS)	96.15%	0.939	0.940	0.952	0.929

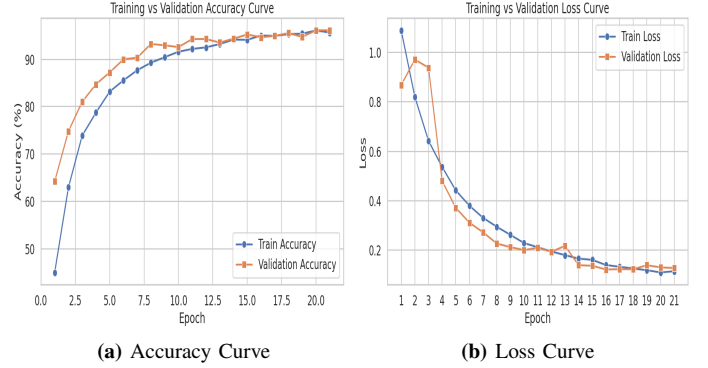


Fig. 3: Training and Validation curves

D. Curve analysis for the proposed model

Fig. 3b and Fig. 3a show the training and validation curves of the proposed model. The loss decreases rapidly and stabilizes near 0.12, while accuracy steadily improves, reaching about 95% for training and 96% for validation. The close overlap of the curves indicates good generalization with minimal overfitting.

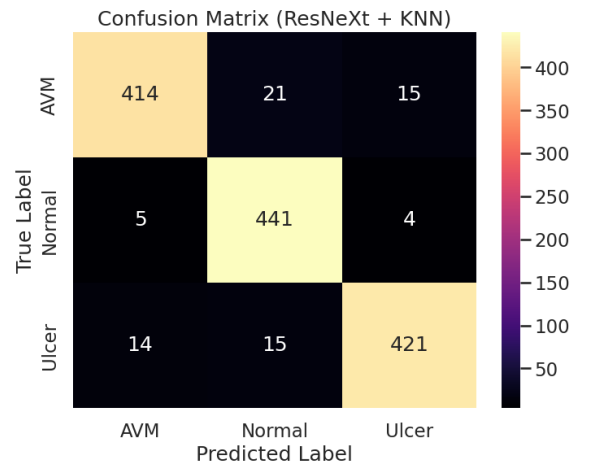


Fig. 4: Confusion Matrix

Fig. 4 presents the confusion matrix of the proposed ResNeXt–KNN model, showing reliable performance across AVM, Normal, and Ulcer. The model correctly classified

414 AVM, 441 Normal, and 421 Ulcer samples, with only minor misclassifications. Recall values of 92%, 98%, and 94% confirm balanced sensitivity, while the strong diagonal dominance underscores the model’s robustness.

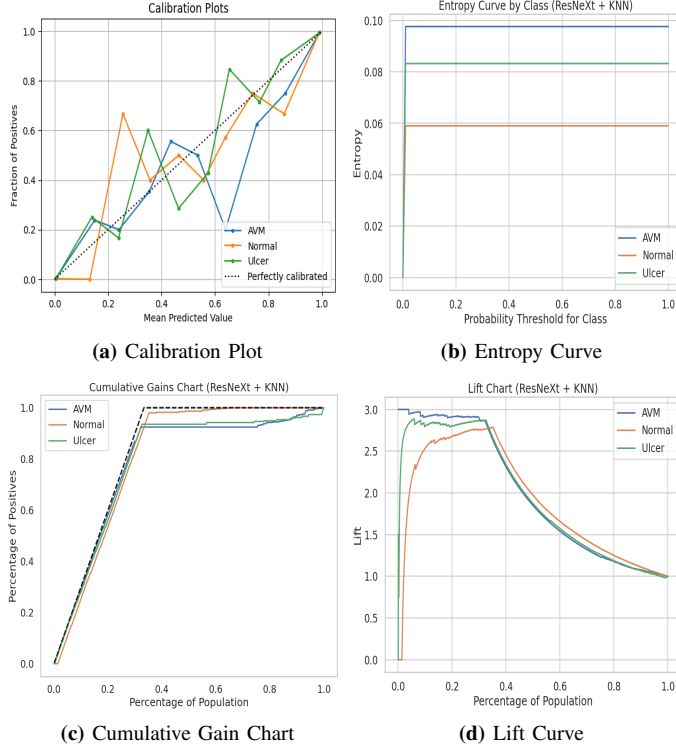


Fig. 5: Evaluation curves: Calibration, Entropy, Gain, and Lift

Fig. 5 summarizes the model’s reliability and ranking behavior. The Calibration Plot (Fig. 5a) shows all classes aligning closely with the ideal curve, with only slight mid-range deviations. The Entropy Curve (Fig. 5b) reflects low uncertainty. The Cumulative Gains Chart (Fig. 5c) captures most positives within the top 20–30%, and the Lift Chart (Fig. 5d) demonstrates strong early lift above 2.5.

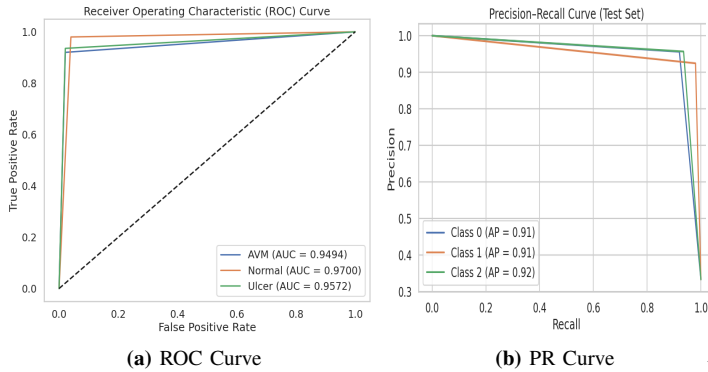


Fig. 6: Evaluation curves: ROC and PR

Fig. 6 illustrates the model’s robustness. The ROC curves

(Fig. 6a) achieve strong AUC scores of 0.9494 (AVM), 0.9700 (Normal), and 0.9572 (Ulcer), showing excellent class separability. The Precision–Recall curves (Fig. 6b) also maintain high precision, with average precision values of 0.91 for AVM and Normal, and 0.92 for Ulcer, confirming reliable performance across thresholds.

E. Comparison with Related studies

Building on prior work in WCE image analysis, the proposed model achieves 96.15% accuracy, placing it among the strongest recent approaches. As shown in Table VII, it outperforms most existing methods, including YOLO-V8 [6], SAVE [7], and MCE [10], while maintaining interpretability and dataset flexibility. This demonstrates the robustness and clinical potential of the ResNeXt with KNN (VCS) framework for automated WCE diagnostics.

TABLE VII: Concise Comparative Analysis of Recent Studies

Author	Method	Results
Sahafi et al. (2024) [6]	YOLO-V8	93.9%
Wang et al. (2024) [7]	Spectrum Aided Vision Enhancer	91% (SSIM)
Bai et al. (2022) [8]	Transformer (spatial pooling)	79.1%
Varam et al. (2023) [9]	ViT	97%
Kim & Chun (2021) [10]	MCE	93.4%
Proposed (2025)	Model ResNeXt–KNN (VCS)	96.15%

F. Explainable AI Results

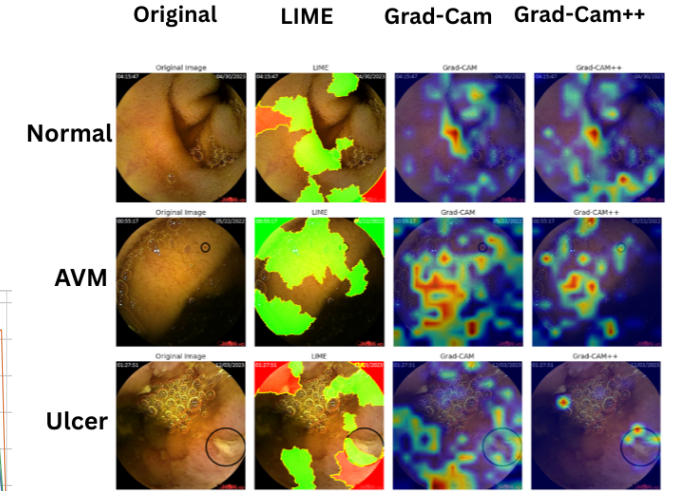


Fig. 7: Explainable AI: GradCAM

Fig. 7 presents interpretability outputs for the ResNeXt with KNN model using LIME, Grad-CAM, and Grad-CAM++. Normal samples show diffuse highlights, while AVM and Ulcer cases exhibit strong activations around vascular and ulcerated regions. LIME highlights segmented patches, and

Grad-CAM/Grad-CAM++ focus on clinically relevant areas, confirming that predictions rely on meaningful pathological cues.

V. CONCLUSION

This study introduced a ResNeXt-50 with KNN framework optimized using the Virus Colony Search (VCS) metaheuristic for automated capsule endoscopy image classification. Combining strong ResNeXt feature extraction with KNN and adaptive parameter tuning, the model achieved 96.15% validation accuracy, a Cohen's Kappa of 0.939, and a Macro F1-score of 0.95. Comparative and ablation studies showed clear improvements over CNN, Transformer-based, and traditional classifiers, while Grad-CAM, Grad-CAM++, and LIME consistently highlighted lesion-relevant regions, supporting interpretability and clinical reliability.

Future work may include expanding the dataset with multi-institutional sources, adding patient-level validation to prevent data leakage, and extending the system to more gastrointestinal conditions. Further optimization for lightweight, real-time deployment and stronger explainable AI methods can enhance interpretability and support broader clinical adoption.

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